

Simulated Annealing

Constraint Satisfaction Problem

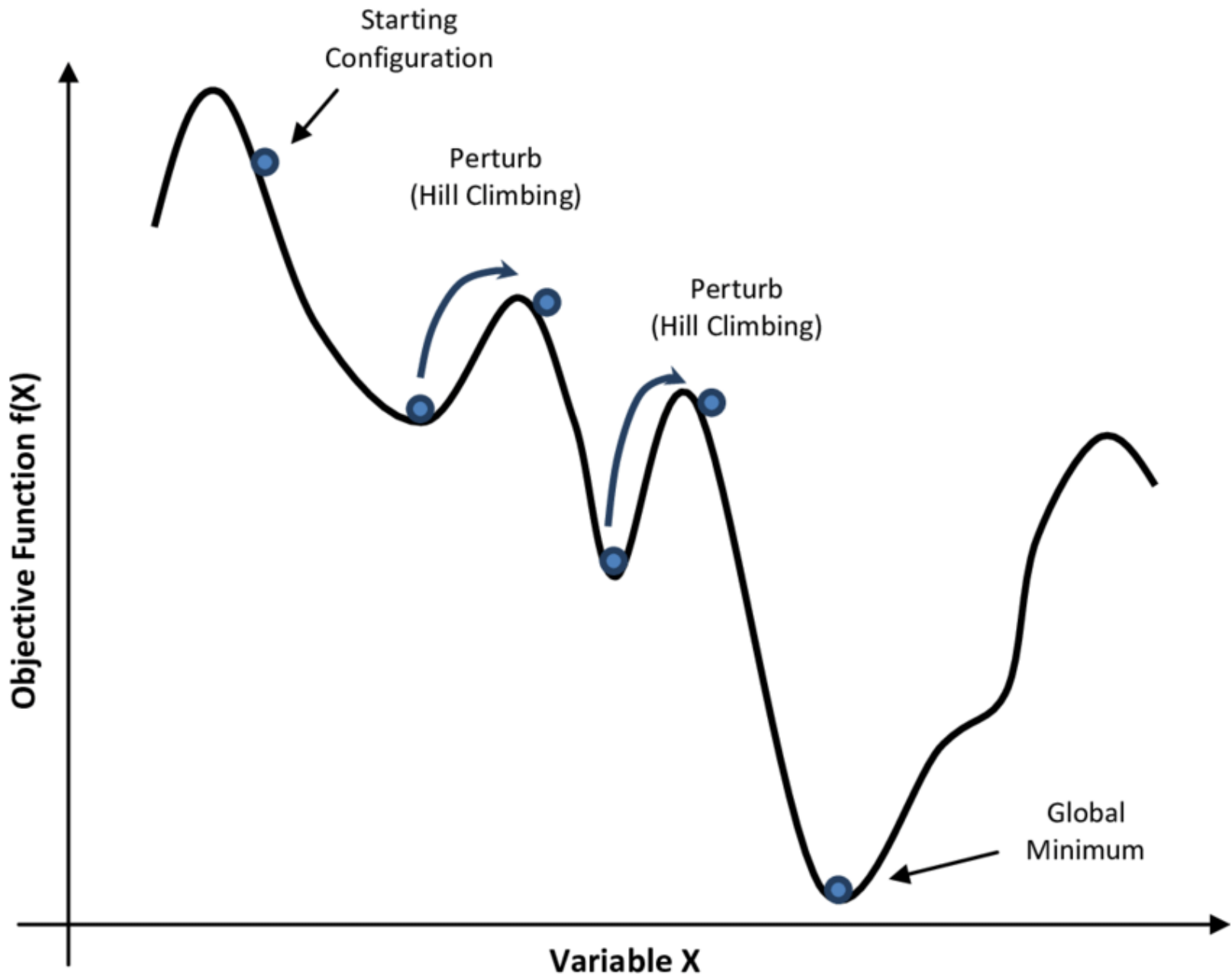
8-puzzle problem

Cryptarithmic Problem

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Simulated Annealing (SA)

- It is a probabilistic optimization technique inspired by the **annealing process** in metallurgy where a material is heated and then slowly cooled to **remove defects and thereby optimize its structure**.
- Simulated Annealing algorithm searches for **an optimal solution in a large search space**, where finding the global optimum is challenging due to the presence of many local optima.



Concepts

State: represents a possible solution.

Energy: Each state has an associated "energy" (or cost) that indicates how good or bad the solution is (lower the energy better).

Temperature: used as a parameter (similar to temperature in annealing).

- At high temperature (higher energy), the **SA algorithm** is likely to get worse solution therefore it explores the search space more broadly.
- As the temperature decreases, the **SA algorithm** tries to move towards the best solution found so far.

Cooling Schedule: It determines how the temperature is lowered over time. A common cooling schedule is exponential, where the temperature is reduced by a constant factor at each step. The schedule is crucial as it balances exploration (searching widely) and exploitation (refining a good solution).

Acceptance Probability: Even if a new state has a higher energy (i.e., it is a worse solution), the algorithm may still accept it based on a probability that depends on the temperature and the difference in energy between the current and new states. As the temperature decreases, this probability becomes smaller, making it less likely to accept worse solutions.

This probability is given by the formula:

$$P(\text{accept}) = \exp\left(-\frac{\Delta E}{T}\right)$$

where ΔE is the change in energy and T is the current temperature.

Procedure:

1. Initialization:

Start with an initial state (random or heuristic-based) and an initial temperature.

2. Iteration:

- Perturb the current state to generate a new neighboring state.
- Calculate the energy of the new state.
- Decide whether to move to the new state based on the acceptance probability.
- Gradually lower the temperature according to the cooling schedule.

3. Termination:

The algorithm terminates when the system has "cooled" sufficiently, i.e., the temperature is very low, or after a fixed number of iterations.

4. Output:

- The final state is the best solution found during the process.

Algorithm:

Let $S = S_0$

For $k = 0$ to $k_{\max}$

$T \leftarrow \text{temperature}(1 - (k+1)/k_{\max})$

$s_{\text{new}} \leftarrow \text{neighbour}(s)$ // Pick a random neighbour

If $P(E(s), E(s_{\text{new}}), T) \geq \text{random}(0, 1)$

$s \leftarrow s_{\text{new}}$

Output: the final state S

One has to provide the following parameters:

1. the state space
2. the energy (goal) function $E()$,
3. the candidate generator procedure $\text{neighbor}()$,
4. the acceptance probability function $P()$
5. the annealing schedule $\text{temperature}()$
6. initial temperature init_temp.

Advantages :

- SA is simple, flexible, and can escape local optima, making it effective for complex optimization problems.
- It is a powerful technique for finding near-optimal solutions in difficult optimization problems where traditional methods might get stuck in suboptimal solutions.

Disadvantages:

The performance of SA heavily depends on the cooling schedule and other parameters, and finding the right settings can be challenging.

Constraint Satisfaction Problem

- A Constraint Satisfaction Problem (CSP) in AI is a mathematical problem defined by a set of objects whose state must satisfy a number of constraints/limitations.
- Applications of CSP: scheduling, planning, resource allocation, etc.

Applications of CSP

- **Scheduling:** Assigning tasks to time slots, ensuring that no two tasks overlap and that resource constraints are respected.
- **Timetabling:** Creating schedules for exams, classes, or meetings where no two events overlap and all constraints (e.g., room availability, instructor availability) are met.

- **Sudoku:** Filling the grid while ensuring each row, column, and 3x3 block contains all digits 1-9 without repetition.

5	3			7				
6			1	9	5			
	9	8					6	
8				6				3
4			8		3			1
7				2				6
	6					2	8	
			4	1	9			5
				8			7	9

5	3	4	6	7	8	9	1	2
6	7	2	1	9	5	3	4	8
1	9	8	3	4	2	5	6	7
8	5	9	7	6	1	4	2	3
4	2	6	8	5	3	7	9	1
7	1	3	9	2	4	8	5	6
9	6	1	5	3	7	2	8	4
2	8	7	4	1	9	6	3	5
3	4	5	2	8	6	1	7	9

Components of a CSP

- **Variables:**

A CSP consists of a set of variables, $X_1, X_2, \dots, X_n$

- **Domains:**

Each variable has a domain, which is the set of possible values it can take.

Ex: if a variable represents a color, its domain might be {red, green, blue, yellow}.

- **Constraints:**

Constraints are rules that limit the values that variables can simultaneously take. They define which combinations of values are allowed or disallowed.

Objective of CSP

- The goal of solving a CSP is to find an assignment of values to all variables that satisfies all the constraints.
- An assignment is **complete** if every variable is assigned a value, and it is **consistent** if it does not violate any constraints.

Map Coloring Problem

- **Variables:** The regions on a map (All states of India)
- **Domains:** Each region can be colored with one of three colors {Red, Green, Blue}.
- **Constraints:** Adjacent regions must not have the same color

The task is to assign a color to each region such that no two adjacent regions have the same color.



Techniques to solve CSP

- **Backtracking:**

A depth-first search approach that assigns values to variables one by one. If a variable assignment violates a constraint, the algorithm backtracks to the previous variable and tries a different value.

- **Constraint Propagation:**

Reduce the search space by eliminating values from the domains that cannot be part of any valid solution.

- **Heuristics:**

- **Most Constrained Variable (MCV):** Choose the variable with the fewest legal values left.
- **Least Constraining Value (LCV):** Choose the value that leaves the most options open for other variables.
- **Degree Heuristic:** Choose the variable that imposes the most constraints on other variables.

- **Local Search:**

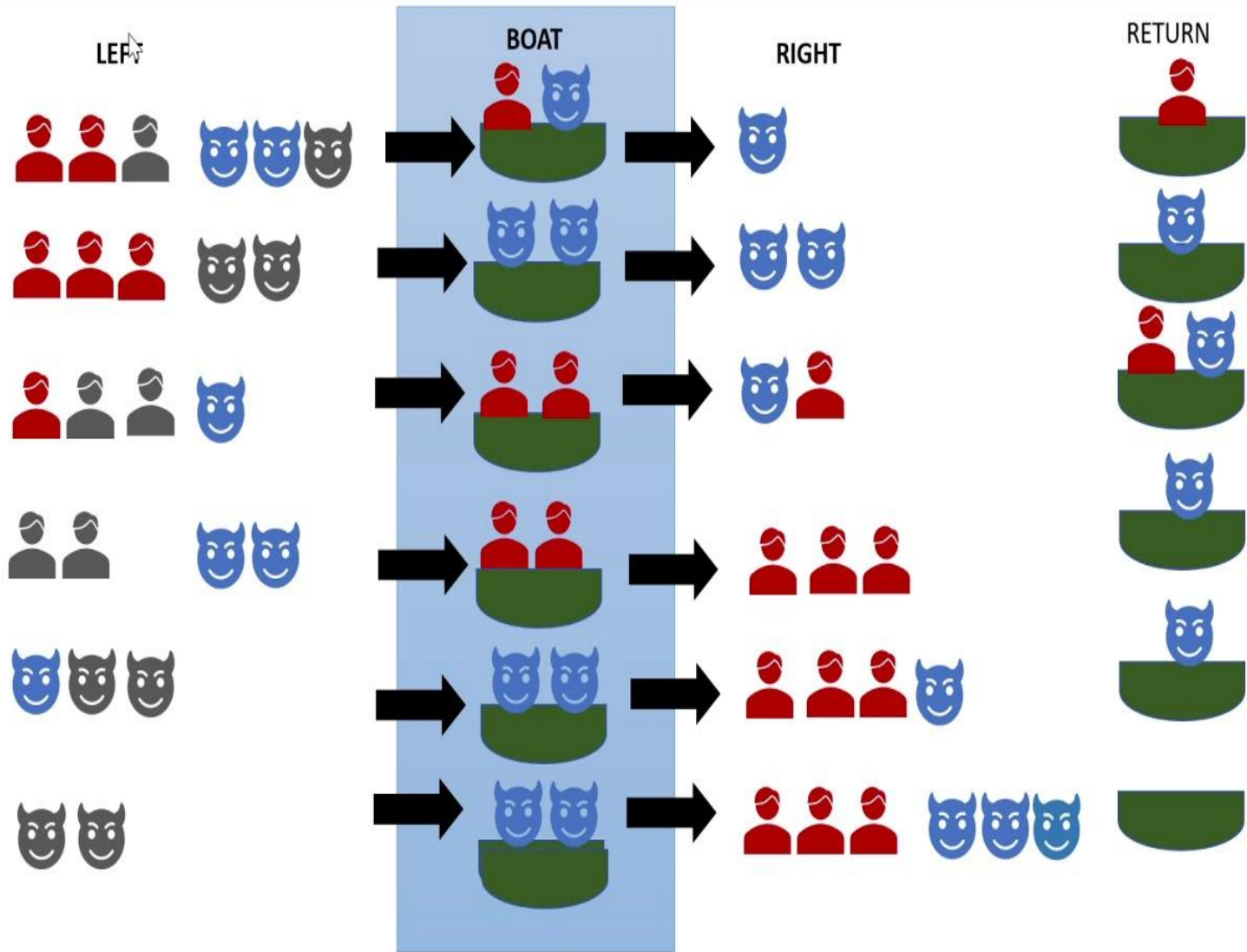
Min-Conflicts: start with an initial assignment and iteratively make small changes to reduce the number of constraint violations.

Missionaries and Cannibals Problem

It is a constraint satisfaction Problem.

The problem

3 missionaries and 3 cannibals must cross a river using a boat which can carry at most 2 people, with the constraint that, at any point of time, if there are more number of cannibals than the missionaries on any side of the river bank, then the cannibals would eat the missionaries. Further, the boat cannot cross the river by itself with no people on board.



Video links for **Missionaries and Cannibals**

https://youtu.be/laLS8gHzROg?si=0_IGQC_hoGR6RScrr



Missionaries & Cannibals game solution.mp4

<https://youtu.be/uyuhcNdb7DI>



River Crossing - 3 Cannibals & 3 Missionaries.mp4

Cryptarithmic problem

It is a type of mathematical puzzle where the digits in a mathematical equation are replaced by letters of the alphabet. The goal is to find the unique digit each letter represents so that the equation holds true.

Key concepts

- **Variables:**

Each letter in the puzzle represents a different digit from 0 to 9.

- **Constraints:**

- No two letters can represent the same digit.
- The solution must satisfy the given arithmetic equation.
- The leading digit of any multi-digit number cannot be zero.

- **Objective:**

The task is to substitute each letter with a digit such that the resulting numerical expression is valid.

SEND+MORE=MONEY

- $S = 9$
- $E = 5$
- $N = 6$
- $D = 7$
- $M = 1$
- $O = 0$
- $R = 8$
- $Y = 2$

S END	9 5 6 7
+ M ORE	1 0 8 5
.....	
M ONEY	1 0 6 5 2

Method

- **Backtracking:** Systematically assign values to each letter and check if the equation is satisfied. If a contradiction arises, backtrack to the previous step.
- **Heuristics:** Apply strategies such as choosing the most constrained variable first or attempting the most promising digits early.

Solution Process

- Let $S=9$ (start with the largest digit) then M has to be 1, so that $S+M=10$
- Next, let $E=8$, (next largest digit). Place 8 wherever E appears.
- Try with 7,6,5 in that order and fix the value of E (here $E=5$)
- Next try with $D=8,7,6,4,3,2$ etc (here $D=7$)
- Try with N =values other than 9,1,5. (here $N=6$)
- Backtrack wherever there is a conflict & try with the next value.

Pseudo code

*Start by examining the rightmost digit of the topmost row, with a carry of 0
If we are beyond the leftmost digit of the puzzle, return true if no carry, false otherwise*

If we are currently trying to assign a char in one of the addends

If char already assigned, just recur on row beneath this one, adding value into sum

If not assigned, then

for (every possible choice among the digits not in use)

make that choice and then on row beneath this one, if successful, return true

if !successful, unmake assignment and try another digit

return false if no assignment worked to trigger backtracking

Else if trying to assign a char in the sum

If char assigned & matches correct,

recur on next column to the left with carry, if success return true

If char assigned & doesn't match, return false

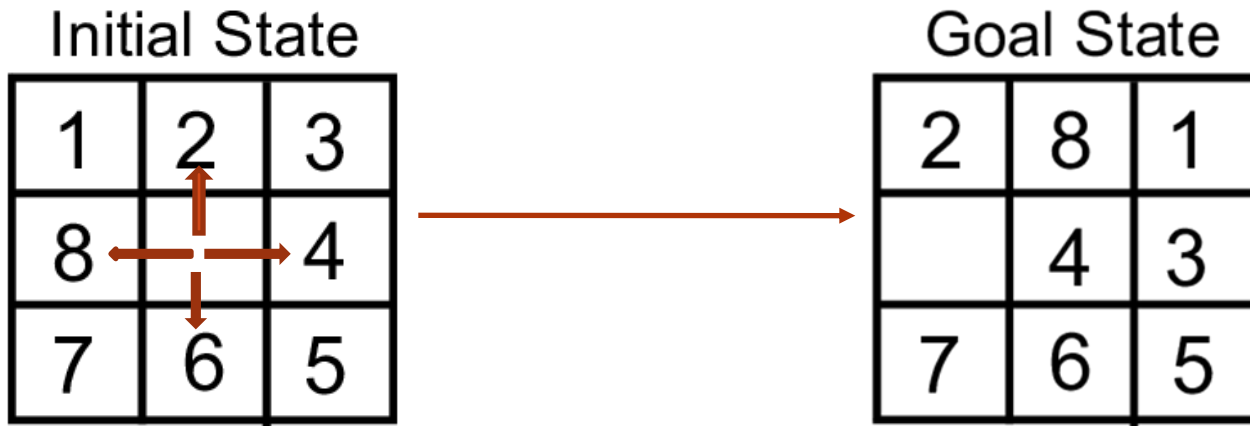
If char unassigned & correct digit already used, return false

If char unassigned & correct digit unused,

assign it and recur on next column to left with carry, if success return true

return false to trigger backtracking

8 Puzzle problem



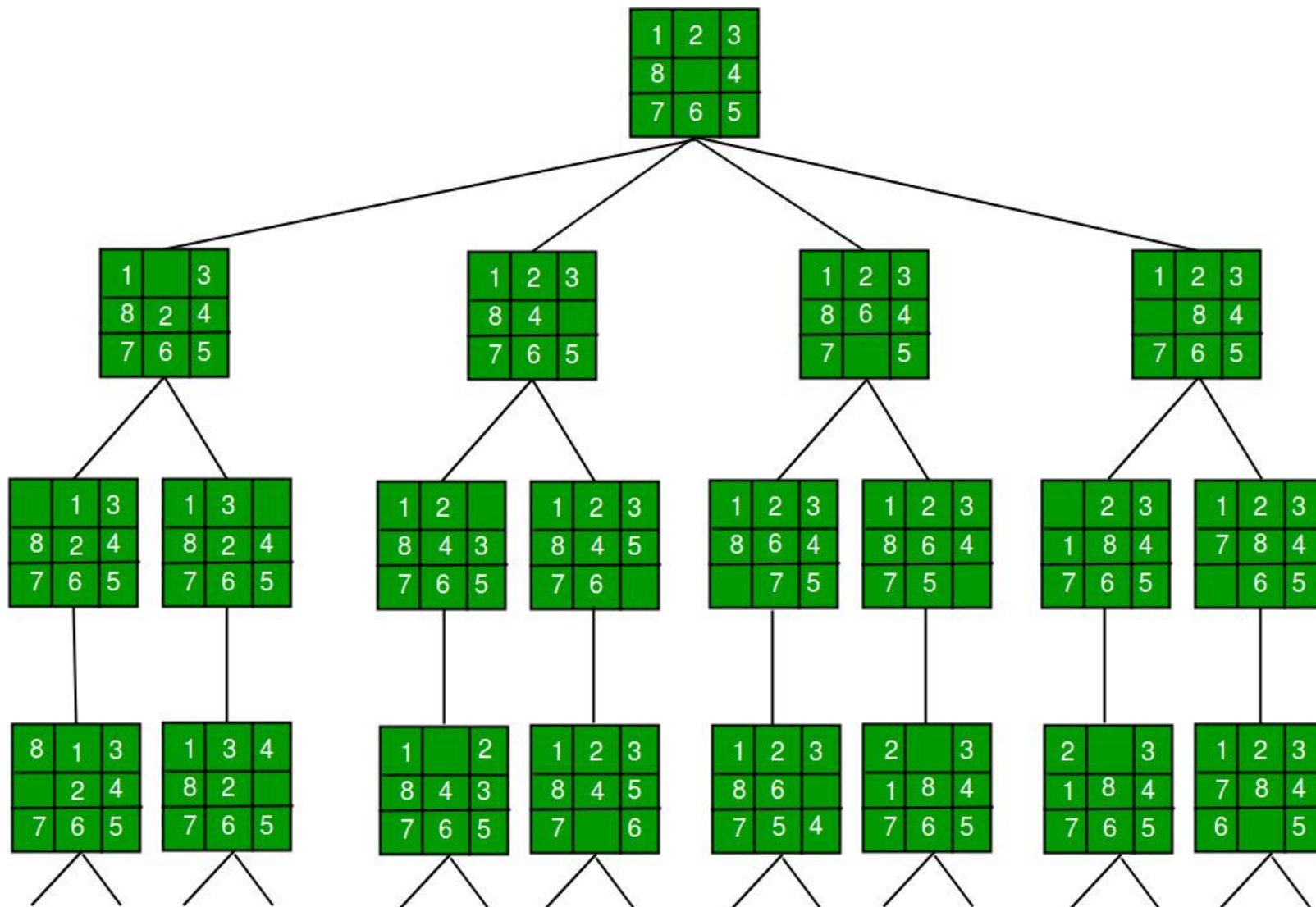
Operators:

Move UP

Move Down

Move Left

Move Right



IS

1	2	3
	4	6
7	5	8

$g=0, h=3, f=g+h=3$

3 steps to reach GS from here

$g=1, h=4, f=5$

	2	3
1	4	6
7	5	8

$g=1, h=2, f=3$

1	2	3
4		6
7	5	8

$g=1, h=4, f=5$

1	2	3
7	4	6
	5	8

4 steps to reach GS from here as one has to go back to IS & then follow the right path.

$g=2, h=1, f=3$

1	2	3
4	5	6
7		8

$g=2, h=3, f=5$

1	2	3
4	6	
7	5	8

$g=2, h=3, f=5$

1		3
4	2	6
7	5	8

$g=3, h=2, f=5$

1	2	3
4	5	6
	7	8

$g=3, h=0, f=3$

1	2	3
4	5	6
7	8	

GS